DATA WAREHOUSING AND DATA MINING
Assignment Answers – Activities 1 to 4

Marks: 50 | Duration: 1 hour

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Activity 1: Design a Star Schema (15 Marks)

Question 1. A college wants to analyze student academic performance across departments, courses, and semesters. A star schema stores measurable academic results in one central fact table and descriptive information in surrounding dimension tables.

Fact table: Fact_Academic_Performance

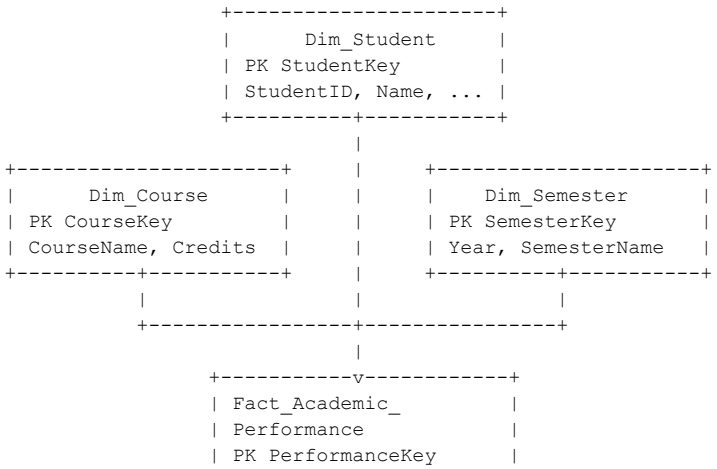
Attribute	Key / Meaning
PerformanceKey (PK)	Unique row identifier for each student-course-semester result
StudentKey (FK)	References Dim_Student.StudentKey
CourseKey (FK)	References Dim_Course.CourseKey
DepartmentKey (FK)	References Dim_Department.DepartmentKey
SemesterKey (FK)	References Dim_Semester.SemesterKey
TotalMarks	Total marks obtained
MaximumMarks	Maximum possible marks
Percentage	TotalMarks / MaximumMarks × 100
AttendancePercentage	Attendance percentage for the course/semester
CreditsEarned	Credits earned
PassCount	1 if passed, otherwise 0

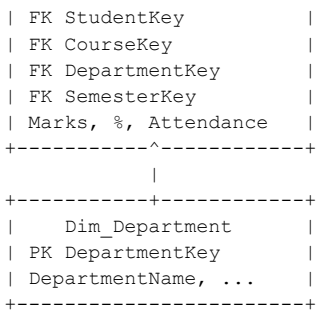
Dimension tables

Dimension table	Attributes (PK shown first)
Dim_Student	StudentKey (PK), StudentID (business key), StudentName, Gender, DateOfBirth, AdmissionYear
Dim_Course	CourseKey (PK), CourseID, CourseName, CourseType, Credits
Dim_Department	DepartmentKey (PK), DepartmentID, DepartmentName, FacultyName
Dim_Semester	SemesterKey (PK), SemesterName, SemesterNumber, AcademicYear, StartDate, EndDate

Relationships: Each dimension's primary key is referenced by the matching foreign key in Fact_Academic_Performance. Each dimension can relate to many fact rows (1-to-many). The grain is one student's result for one course in one semester.

Star schema diagram




| FK StudentKey |
| FK CourseKey |
| FK DepartmentKey |
| FK SemesterKey |
| Marks, %, Attendance |
Dim_Department
| PK DepartmentKey |
| DepartmentName, ... |

Activity 2: Perform ETL Operations (10 Marks)

Question 2. ETL integrates data from admission, examination, and attendance systems into a consistent warehouse format.

1. Extract

Read records from source database tables or CSV exports. Preserve source identifiers and extraction date for traceability.

Sample raw input records

Source	ID	Name	Department	Date	Value
Admission	S101	Asha	CSE-AI	15/06/2024	Year=2024
Exam	S101	Asha	Computer Science & AI	2024-11-20	Marks=86
Attendance	S101	Asha	CSE AI	20-11-2024	Attendance=92%
Exam	S102	Ravi	CSE	2024/11/20	Marks=missing
Exam duplicate	S101	Asha	CSE-AI	20-11-2024	Marks=86

2. Transform

- Standardize department aliases: “CSE-AI”, “CSE AI”, and “Computer Science & AI” → “CSE–Artificial Intelligence”.
- Convert dates to ISO format (YYYY-MM-DD).
- Remove duplicate rows using a business key such as StudentID + source event + date; retain one valid record.
- Handle missing marks by keeping NULL and flagging for validation; do not silently replace unknown marks with zero.
- Trim whitespace, standardize case, validate ranges (marks 0–100; attendance 0–100), and verify student IDs against the master student list.

3. Load

Load cleaned data into a staging/target table in a controlled batch. Use StudentID as a business key and upsert logic to prevent duplicate loads. Send rejected records to a data-quality log for correction.

Final cleaned target table (illustrative)

StudentID	StudentName	Department	EventDate	Marks	AttendancePct	QualityNote
S101	Asha	CSE–Artificial Intelligence	2024-11-20	86	92	Duplicate removed; dates and department standardized
S102	Ravi	Computer Science	2024-11-20	NULL	NULL	Missing marks retained as NULL; validation required

ETL flow: Source systems → Extract → Staging area → Clean / standardize / deduplicate / validate → Load into warehouse tables → Data-quality log and reports.

Activity 3: Demonstrate OLAP Operations (15 Marks)

Question 3. Sample sales data; each row represents monthly sales for one product in one location.

Year	Month	Product	Location	SalesAmount (₹)
2025	Jan	Laptop	Chennai	1000
2025	Feb	Laptop	Chennai	1200
2025	Jan	Phone	Chennai	800
2025	Feb	Phone	Chennai	900
2025	Jan	Laptop	Nellore	700
2025	Feb	Laptop	Nellore	600
2025	Jan	Phone	Nellore	500
2025	Feb	Phone	Nellore	550
2026	Jan	Laptop	Chennai	1300
2026	Feb	Laptop	Chennai	1400
2026	Jan	Phone	Chennai	950
2026	Feb	Phone	Chennai	1000

a) Roll-up – monthly sales summarized into yearly sales

Roll-up groups month-level data into year-level totals across all products and locations.

Year	Total annual sales (₹)
2025	6,250
2026	4,650

b) Drill-down – yearly sales displayed month-wise

Year	Month	Total sales (₹)
2025	Jan	3,000
2025	Feb	3,250
2026	Jan	2,250
2026	Feb	2,400

Drill-down moves from a summarized year to the more detailed month level.

c) Slice – retrieve sales for one product (Laptop)

Year	Month	Location	Sales (₹)
2025	Jan	Chennai	1,000
2025	Feb	Chennai	1,200
2025	Jan	Nellore	700
2025	Feb	Nellore	600
2026	Jan	Chennai	1,300
2026	Feb	Chennai	1,400

Slice fixes one dimension value: Product = Laptop. Only Laptop records are included.

Activity 3: OLAP Operations (continued)

d) Dice – selected products, locations, and time periods

Condition: Product in {Laptop, Phone}, Location = Chennai, Year = 2025. This returns a sub-cube restricted by multiple dimension filters.

Year	Month	Product	Location	Sales (₹)
2025	Jan	Laptop	Chennai	1,000
2025	Feb	Laptop	Chennai	1,200
2025	Jan	Phone	Chennai	800
2025	Feb	Phone	Chennai	900

e) Pivot – compare products and locations

Pivot table for 2025 total sales, with products as rows and locations as columns:

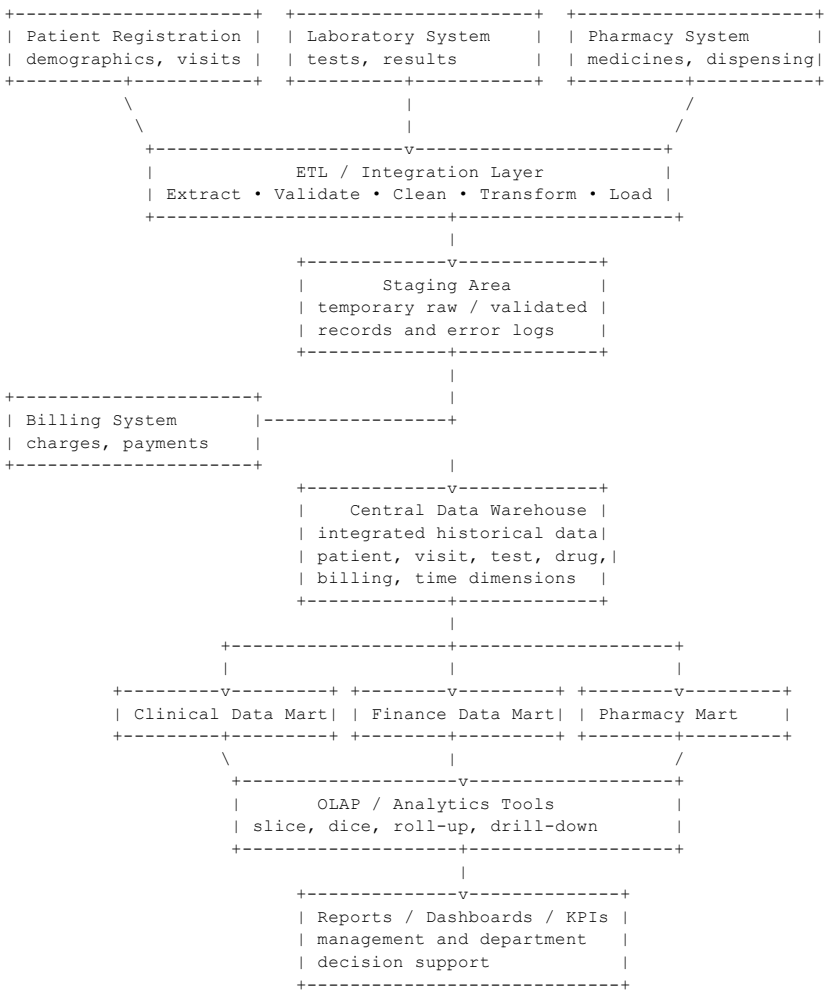
Product	Chennai (₹)	Nellore (₹)	Grand Total (₹)
Laptop	2,200	1,300	3,500
Phone	1,700	1,050	2,750
Grand Total	3,900	2,350	6,250

Pivot rotates the view of the same data without changing the underlying values. OLAP helps managers compare results at different levels and across dimensions.

Activity 4: Draw Data Warehouse Architecture (10 Marks)

Question 4. A hospital data warehouse integrates registration, laboratory, pharmacy, and billing data for reliable operational and management decisions.

Labelled hospital data warehouse architecture



Functions of each component

Component	Function
Data sources	Operational systems that capture patient details, laboratory results, pharmacy transactions, and billing information.
ETL / integration	Extracts source data, standardizes codes and dates, checks quality, resolves duplicates, and maps fields to warehouse structures.
Staging area	Temporary workspace for extracted data and transformations; isolates invalid records before loading.
Central data warehouse	Stores integrated, consistent, historical data for analysis rather than day-to-day transaction processing.
Data marts	Subject-focused subsets for clinical quality, finance/revenue, and pharmacy/inventory analysis.
OLAP tools	Enable multidimensional analysis, comparisons, summaries, and drill-down into detailed data.
Reports and dashboards	Present KPIs such as patient volume, lab turnaround time, medicine usage, revenue, unpaid bills, and department performance.

How it supports decision-making

The architecture gives managers a consistent view across departments and over time. They can identify high-demand services, monitor laboratory turnaround time, forecast medicine needs, analyze revenue and outstanding bills, compare department performance, and allocate staff or resources based on evidence. Role-based access, audit trails, data validation, and protection of patient information should be applied throughout.

— End of Assignment —